

System Architecture & Operational Guide

How Panchranga Works at this Point of Time: End-to-End Technical Deep Dive

Status: Production Active	Active Sources: 96 Verified Feeds	Languages: 13 Indian Languages
Editorial Framework: 3 Perspective Lanes	Ingestion Cadence: Every 20 Mins (Cron)	Embeddings: 384-Dim MiniLM-L6 Vector
Database: Supabase Postgres (pgvector)	AI Models: Groq Qwen / Gemini Flash	Web Stack: Next.js 14 App Router

1. Executive Summary & Core Mission

Panchranga (पंचरंग) is an open-source, automated news intelligence platform designed specifically for the Indian media ecosystem. The platform addresses severe media polarization, echo chambers, and commercial bias by aggregating coverage across the entire spectrum and presenting multiple perspectives side by side on a unified canvas.

The Hindi word Panchranga signifies “many colors” (literally five colors). The platform operates on the core axiom that in a complex pluralistic democracy, no single media outlet captures the entire truth of an event. Panchranga enforces a strict editorial charter: No human editors, no paywalls, no partisan bias labels, and complete transparency.

Pillar	Implementation Mechanism	Democratic Value
Automated Aggregation	96 RSS, Reddit, and API feeds polled every 20 minutes via bounded asynchronous workers.	Removes selective curation and algorithmic echo chambers.
Three-Lane Separation	Strict partitioning into Mainstream, Grassroots, and Public Discourse lanes.	Exposes narrative divergence and institutional vs ground reality.
Semantic Clustering	384-dimensional vector embeddings with cosine similarity grouping (threshold 0.45).	Connects identical events regardless of language or vocabulary.
Guardrailed AI Briefs	Cascading LLMs generate 35-word neutral summaries; sensitive topics auto-bypass.	Delivers instant, objective context without sensationalist framing.

The Panchranga Philosophy: Why Single-Perspective Feeds Fail in India

India's linguistic and regional diversity means stories unfold differently across national broadsheets, regional language outlets, and civic discussion spaces. A national daily may emphasize government economic data, an independent rural reporter highlights local land acquisition issues, while citizen forums debate immediate real-world fallout. Panchranga brings all three dimensions onto a single screen so readers see the complete truth.

2. End-to-End System Architecture & Data Flow

Panchranga is constructed as an asynchronous, event-driven pipeline running on scheduled cron jobs, paired with a modern Next.js 14 serverless web application. The lifecycle of a news item from external ingestion to browser presentation follows six deterministic stages:

Stage 1: Multi-Source Ingestion scripts/fetch-rss.ts	Pulls from 96 feeds across 4 lanes (58 Mainstream, 22 Grassroots, 9 Discourse, 7 Aggregators) across 13 languages. Concurrency bounded by p-limit(7). Cloudflare-protected feeds routed via Google News syndication fallbacks.
Stage 2: Aggregator Deduplication lib/dedup.ts	Pooled aggregator feeds (Currents API + Google News RSS) are checked against direct feeds within a ±2h publication window using exact URL, normalized title, and token Jaccard similarity (>= 85%) to prevent redundant story counts.
Stage 3: Vector Embedding lib/embeddings.ts	Raw headlines and summaries are stripped of stopwords and projected into a 384-dimensional vector space using multilingual token hashing and sub-word 3-grams, normalized via L2 unit length for cosine dot-product calculations.
Stage 4: Topic Hub Clustering lib/clustering.ts	Single-pass clustering compares new items against active hubs in a rolling 48-hour window. Articles with Cosine Similarity >= 0.45 join the highest-matching cluster; novel events spawn new Topic Hubs. Lane counts are recalculated dynamically.
Stage 5: Guardrailed AI Synthesis lib/summarizer.ts	Multi-model cascade (Groq Qwen 2.5 32B -> Gemini Flash -> OpenRouter -> Article Snippet Rule). Generates strict 1-2 sentence neutral facts (max 35 words). High-stakes communal/sub-judice keywords trigger automated bypass (Sources-Only Mode).
Stage 6: Dual Persistence & UI Next.js 14 App Router	Dual storage layer: Supabase PostgreSQL (pgvector) + local JSON files (data/topic-hubs.json). Rendered via Next.js SSR/ISR with 3-column layout, Coverage Bars, mobile tabs, live fact-checking ticker, and daily morning newsletter.

3. The Three-Lane Architecture & Source Registry

Panchranga isolates coverage into distinct editorial lanes to reveal media framing and narrative divergence:

Media Lane	Color & UI Code	Source Count	Characteristics & Editorial Mandate
Mainstream Lane 1	Blue (#2563EB)	58 Sources (60.4%)	National English & Hindi daily broadsheets, television networks, wire agencies, official government press releases (Indian Express, Times of India, NDTV, BBC India, Reuters, PIB, Dainik Bhaskar). Focus: Institutional record, press statements, national distribution.
Grassroots Lane 2	Green (#16A34A)	22 Sources (22.9%)	Independent newsrooms, investigative collectives, and regional language ground reporters (The Wire, Scroll.in, Newslaundry, The News Minute, Article 14, The Mooknayak, PARI, Khabar Lahariya, Down To Earth). Focus: Ground impact, rural distress, human rights, marginalized communities.
Public Discourse Lane 3	Amber (#D97706)	9 Sources (9.4%)	Citizen journalism networks, online civic discussion threads, active Indian subreddits (r/india, r/IndiaSpeaks, r/delhi, r/bangalore, r/mumbai, Youth Ki Awaaz, Sabrang India). Focus: Unfiltered citizen sentiment, public debate, ground-level feedback.
Aggregator Auxiliary	Gray (#64748B)	7 Sources (7.3%)	Multi-publisher pooled syndication streams: Currents API (with 250 req/day quota guard) and Google News RSS editions (English, Hindi, Punjabi, Urdu, Odia, Assamese). Undergoes deduplication before assignment.

Language Breakdown across 13 Indian Languages: English (55), Hindi (12), Punjabi (4), Odia (4), Tamil (3), Urdu (3), Assamese (3), Telugu (2), Kannada (2), Malayalam (2), Marathi (2), Gujarati (2), Bengali (2).

4. Data Ingestion Pipeline & Anti-WAF Defense

The data ingestion pipeline (scripts/fetch-rss.ts) is orchestrated every 20 minutes via GitHub Actions. It is engineered with robust defensive mechanisms against bot blocks, network failures, and publisher variance:

- **Bounded Concurrency:** Feeds are ingested with p-limit(7) concurrency. This avoids socket starvation while fetching up to 30 items per feed across 96 sources within ~45 seconds.
- **Anti-WAF Syndication Fallbacks (FEED_FALLBACKS):** Major publishers (BBC, Livemint, Hindustan Times, Scroll.in) employ Cloudflare bot-protection that blocks automated crawlers with HTTP 403. Panchranga automatically reroutes failing feeds through Google News RSS syndication queries (e.g. site:bbc.com/news/world/asia/india), maintaining 100% feed availability without headless browsers.
- **Reddit Pacing & Multi-Subreddit Ingestion:** Subreddits are fetched with polite 2,000ms–5,000ms pacing delays, with fallback to unified multi-subreddit RSS (r/india+IndiaSpeaks+delhi+bangalore+mumbai/.rss). This guarantees complete civic discourse coverage without hitting Reddit 429 rate limits.
- **Strict 2.5s OpenGraph Extraction:** The engine runs live HTML scraping only for the top 2 articles per source missing RSS enclosure thumbnails, enforcing a strict 2.5-second abort signal to maintain high pipeline throughput.
- **Source Health Telemetry:** Every cycle updates Supabase sources with last_status ('healthy' / 'error'), last_attempted_at, and last_error for real-time monitoring.

5. Aggregator Deduplication Engine (lib/dedup.ts)

Pooled aggregator feeds (Currents API and Google News RSS) frequently syndicate news stories already ingested from direct publisher RSS feeds. Panchranga uses a multi-tier deduplication algorithm to eliminate duplicate stories before clustering:

Step	Logic & Algorithm	Threshold / Criteria
1. Title Normalization	Strips publisher branding suffixes (- The Hindu, NDTV, — BBC). Cleans punctuation via Unicode-aware regex ([^\p{L}\p{N}\s]) while preserving Hindi/Indic scripts.	Clean lowercased canonical token string.
2. Time-Window Check	Compares candidate article timestamp against existing direct items. Only stories published within the same temporal window are evaluated.	±2 hours (7,200,000 ms)
3. Exact URL Match	Direct canonical URL equality.	Exact string match (100% duplicate)
4. Normalized Title Match	Exact match of sanitized canonical titles.	Exact string match (100% duplicate)
5. Token Jaccard Overlap	Word-level Jaccard similarity: Set(A) ∩ Set(B) / Set(A) ∪ Set(B) on words with length > 2.	Similarity >= 0.85 (85% token overlap)

Production Deduplication Metrics

During standard ingestion cycles, the deduplication engine evaluates ~100 aggregator articles against ~900 direct feed items. An average of 15%–25% of aggregator articles are identified as duplicates and filtered out, preventing artificial inflation of mainstream lane counts and keeping topic hub metrics pristine.

6. Vector Embeddings & Event Clustering Engine

Panchranga groups disparate news reports into unified Topic Hubs using vector semantic similarity. Rather than relying on human categorization or fragile keyword matching, the system calculates the geometric proximity of articles in a 384-dimensional space.

Multilingual Embedding Mechanics (`lib/embeddings.ts`):

Each raw item title and snippet is passed through an Indic-aware embedding tokenizer. Stopwords across English and Indian languages (news, latest, india, updates, pm, says, etc.) are purged. The engine computes both full word token hashes and sub-word 3-grams, mapping them into a 384-element float array. The vector is normalized to unit length (L2 norm = 1.0) so that Cosine Similarity is equivalent to the dot product: $\text{dot}(A, B) = (A[i] * B[i])$.

Single-Pass Clustering Algorithm (`lib/clustering.ts`):

1. Rolling 48-Hour Window: During each clustering cycle (`scripts/cluster-items.ts`), the system evaluates raw items from the past 48 hours to maintain fresh, active topic hubs.
2. Hub Comparison: For each incoming raw item, the engine calculates the maximum cosine similarity against all articles inside each existing topic hub.
3. Threshold Association: If `max_similarity` ≥ 0.45 , the item is attached to that Topic Hub. Duplicate URLs within the same hub are automatically rejected.
4. Hub Spawning: If no existing hub satisfies the 0.45 threshold, a new Topic Hub is spawned with a unique UUID, adopting the article's title as the hub headline.
5. Lane Re-indexing: Once clustering completes, each hub's `mainstream_count`, `grassroots_count`, and `discourse_count` are recomputed dynamically.

Clustering Throughput & Performance in Production

In production, clustering 1,000 raw items into ~700 topic hubs (including 100+ multi-source breaking hubs) executes in under 4 seconds. The system syncs cluster IDs to Supabase Postgres in concurrent batches of 50 and persists an instant local cache in `data/topic-hubs.json`.

pgvector Stored Procedure (`match_topic_hubs`):

Supabase performs cosine distance vector matching directly in PostgreSQL via the PL/pgSQL function:

```
1 - (raw_items.embedding <=> query_embedding) AS similarity
```

This allows real-time vector queries across thousands of items in under 15ms using PostgreSQL IVFFlat / HNSW vector indexing.

7. AI Neutral Summarizer & Safety Guardrails

Panchranga features an automated editorial synthesizer (`lib/summarizer.ts`) that generates objective, 1-2 sentence neutral briefings for multi-source topic hubs. It is engineered with two critical design requirements: failover resilience and strict ethical guardrails.

Priority Tier	Provider & Model	Role & Quota Profile	Failure / Cooldown Behavior
Tier 1 (Primary)	Groq Cloud qwen/qwen3.8-27b	Sub-second inference (400ms). Used for ultra-fast batch and on-demand generation.	On HTTP 429: sets 60s cooldown (1h if daily TPD limit hit) and cascades to Tier 2.
Tier 2 (Failover)	Google Gemini gemini-3.8-flash	High daily token quota (1,500 requests/day, 1M TPM). Deep multilingual reasoning.	On HTTP 429: sets 60s cooldown and cascades to Tier 3.
Tier 3 (Fallback)	OpenRouter Free nemotron-3.5 / ling-3.0	Multi-model open pool. Ensures free tier resilience when primary keys are exhausted.	Cascades to Tier 4 if all free models return errors.
Tier 4 (Rule Engine)	Deterministic Snippet Regex Sentence Extractor	Zero API calls. Extracts the cleanest factual sentence from the lead publisher.	Guarantees 100% uptime with zero empty summary cards.

Safety Bypass Guardrails: Sources-Only Mode

Zero AI Hallucination Policy: When a news hub contains sensitive or legally high-stakes keywords (communal violence, communal riot, mob lynching, hate speech, sub-judice, ongoing trial, curfew imposed), the system immediately bypasses AI generation. The UI renders the verified publisher headline accompanied by an alert: 'AI Summary bypassed for high-stakes sensitive topic. Displaying verified reporting only.'

Deep Article Scraping (`lib/article-extractor.ts`):

Unlike aggregators that summarize only brief RSS snippets, Panchranga uses an asynchronous HTTP client to scrape the primary news article webpage directly. It filters boilerplate (cookie warnings, subscribe banners, copyright notices) and extracts up to 1,500 characters of substantive paragraph text within a 3.5s timeout window. This provides the LLM with concrete facts, casualty figures, and official quotes.

Strict Editorial Prompt Constraints:

The system prompt instructs the model: 'Write a concise, fact-rich 1-2 sentence neutral briefing (max 35 words) capturing key event, concrete facts, numbers, and locations. Plain English only. No source names, no media labels, no quotation marks.'
Post-processing filters strip any accidental thought tags (`<think>`) or boilerplate prefixes.

8. Database Schema & Data Models

Panchranga uses Supabase PostgreSQL with the pgvector extension for high-speed vector similarity lookups. The database schema (scripts/schema.sql and scripts/newsletter_schema.sql) consists of five core tables:

Table Name	Keys & Constraints	Key Columns & Types	Operational Purpose
sources	id uuid PK	name, lane, type, feed_url UNIQUE, language, region, is_active, last_status	Registry of all 96 news feeds with health telemetry and lane tags.
raw_items	id uuid PK source_id FK cluster_id FK	title, url UNIQUE, published_at, raw_summary, og_image, category, embedding vector(384)	Every individual news report ingested, indexed by URL and 384-dim vector embedding.
topic_hubs	id uuid PK	title, ai_summary, first_seen_at, last_updated_at, item_count	Clustered story hubs representing single events with synthesized briefings.
newsletter_subscribers	id uuid PK	email UNIQUE, subscribed_at, is_active, unsubscribe_token uuid	Active subscribers for daily morning email delivery.
newsletter_sends	id uuid PK	sent_at, subscriber_count, hub_ids text[], status	Audit log of dispatched newsletter editions and featured hubs.

9. Frontend Web Application & User Experience

Panchranga’s user interface is built with Next.js 14 App Router, React 18, and Tailwind CSS. The design embraces a high-density, authoritative editorial broadsheet aesthetic inspired by traditional Indian newspaper typography (Georgia serif headlines, clean Inter body text, and uppercase monospace metadata tags).

Component / Page	Route / Path	Key Features & Interactive Capabilities
Three-Column Homepage	/ (app/page.tsx)	- Left (280px): 'Today's Briefing' with micro-thumbnails & coverage bars. - Center: 380px featured Hero Story banner + paginated list layout. - Right (300px): 'Most Covered' stories + Live Fact-Check Ticker (Alt News, BOOM, Newschecker, Factly).
Topic Hub Detail Page	/hub/[id]	- Coverage Bar: Full-width cross-media proportional split visualization. - HubAiOverview: Neutral briefing with typing animation and disclaimer. 3-Lane Side-by-Side: Parallel columns for Mainstream, Grassroots, Discourse. - Embedded Media: Native YouTube broadcast players & Reddit thread embeds.
Sources Registry	/sources	Live public directory of all 96 verified sources with active status indicators, media lane badges, language tags, regions, and live article counts.
Editorial About Page	/about	Full disclosure of editorial charter, algorithm mechanics, fact-checking organization links, GitHub repository, and feedback channels.
Panchranga Custom Loader	PanchrangaLoader.tsx	5-dot horizontal wave animation in brand colors (#4A56E2, #00B8A9, #F5A623, #E84393, #FF6B6B), with smooth wordmark resolution and prefers-reduced-motion accessibility support.
Safe Image Handling	/api/og-image & SafeImage.tsx	Proxies external images to bypass CORS/hotlink blocking; automatically falls back to clean monogram tiles if publisher CDNs return HTTP 403/404.

10. Automation, CI/CD & Daily Newsletter System

Panchranga is architected for autonomous, zero-maintenance operations. No daily human intervention is required to ingest news, cluster hubs, synthesize summaries, or deliver morning newsletters.

GitHub Actions Scheduled Workflows:

1. Pipeline Ingestion Cron (.github/workflows/ingest-cron.yml): Runs every 20 minutes (cron: `*/*20 * * * *`). Checks out code on Node 22, injects Supabase and LLM API secrets, and executes `npm run pipeline` (fetch RSS + cluster items).
2. Daily Newsletter Cron (.github/workflows/newsletter.yml): Triggers every morning at 7:00 AM IST (1:30 AM UTC). Dispatches an authenticated HTTP POST request to `https://panchranga.vercel.app/api/newsletter/send`.

Daily Morning Newsletter Engine (app/api/newsletter/send/route.ts):

- Story Selection: Automatically selects the top 3 most significant topic hubs from the past 24 hours based on source count and cross-media diversity.
- Email Construction: Assembles a responsive, broadsheet-styled HTML email featuring lead imagery, AI-synthesized briefings, participating publisher badges, and direct links to full 3-lane coverage.
- Batch Delivery via Resend API: Reads active subscribers from Supabase and delivers emails in concurrent batches of 50 to respect Resend gateway rate limits.
- 1-Click Unsubscribe: Every email includes a secure cryptographic UUID token allowing instant 1-click unsubscription without requiring a login.

11. Production Metrics & Operational Resilience

Metric Dimension	Production Value	Architectural Safeguard
Total Configured Feeds	96 Sources across 13 languages	Automated fallback to Google News syndication on WAF block.
Ingestion Frequency	Every 20 Minutes (72 cycles / day)	Bounded concurrency <code>p-limit(7)</code> prevents serverless timeouts.
Active Articles Cached	1,000 Articles in rolling window	URL unique constraint prevents duplicate PostgreSQL upserts.
Active Topic Hubs	707 Hubs (107 Multi-Source Hubs)	Orphaned hubs automatically cleaned up during clustering.
AI Summarization	4-Tier Resilient Cascade	Groq -> Gemini -> OpenRouter -> Lead Snippet Rule fallback.
Offline Reliability	100% Zero-Downtime Cache	Local JSON files mirror Supabase for instant fallback.

Conclusion: The Future of Indian News Aggregation

Panchranga demonstrates that algorithmic automation, semantic vector search, and strict ethical guardrails can replace subjective human gatekeepers in modern media. By presenting Mainstream, Grassroots, and Public Discourse side by side, Panchranga equips Indian citizens with the complete spectrum of facts — allowing readers to draw their own informed conclusions. Every Color of the Story.